中熔电气
SINOFUSE

RS0ER-L2Q/L2Q-N

产品规格书

PRODUCT SPECIFICATION

SINOFUSE



成为能量保护系统的集成方案供应商
为整个人类的能量转换活动的安全运行提供保障
Provide integrated solution of energy protection systems and safeguard energy conversion processes for mankind

文件变更履历表/ Revision Record

文件名称/ File name： RS0ER-L2Q/L2Q-N

文件编号/ File No.： ZR/SCG-0050 A0

Version	Content	Implement Date	Author
A0	由产品规格书 ZR/YC-0709转版，内容无变更，项目增加/Transferred from product specification ZR/YC-0709, no change in content, increase in items	2025-4-9	Wangchengyu

产品技术参数/Product technical parameters

●性能参考/ Conform to： IEC 60269-4
●安全性和应用环境可靠性参考/Safety and application environment reliability reference：ISO 8820、GB/T 31465
●额定电压/Rated voltage： DC1000V
●额定电流/ Rated current： 200A～400A
●保护类别/Protection category：aR
●分断能力/ Breaking capacity： 8In-50kA
●符合/ Compliant :RoHS
●认证/Certification: CCC、CE、TUV

NO	Part Number	Size	In (A)	I^2t (A^2s)		Loss (W)	Resistance value mΩ (20℃) ±15%	Weight (g)	Min Package (pcs)	Max Package (pcs)	Remark
				Prearc	Clearing	In					
1	RS0ER-L2Q 200A	R	200	7632	22465	86	0.876	320±10g	6	18	M8 12±1N·m
2	RS0ER-L2Q 250A		250	14958	44031	96	0.626				
3	RS0ER-L2Q 315A		315	30527	89860	110	0.452				
4	RS0ER-L2Q 350A		350	43958	129398	113	0.378				
5	RS0ER-L2Q 400A		400	63300	186333	126	0.322				
6	RS0ER-L2Q-N 200A		200	7632	22465	86	0.876	289±10g	8	24	
7	RS0ER-L2Q-N 250A		250	14958	44031	96	0.626				
8	RS0ER-L2Q-N 315A		315	30527	89860	110	0.452				
9	RS0ER-L2Q-N 350A		350	43958	129398	113	0.378				
10	RS0ER-L2Q-N 400A		400	63300	186333	126	0.322				

注:表中RS0ER-L2Q/ RS0ER-L2Q-N 400A I^2t数据为标准环境下测得典型数据，测试条件为DC1010V、50.2kA 、时间常数为11.5ms。
Note:The RS0ER-L2Q/ RS0ER-L2Q-N 400A I^2t data in the table are typical data measured in a standard environment with DC1010V, 50.2kA, and a time constant of 11.5ms.

产品特性/ Product features

本产品为螺栓连接式熔断器，具有强限流能力、高分断能力等特点，可作为电动汽车驱动电力系统及其他设备的短路保护。
This product is a bolt-on fuse with strong current limiting capability and high breaking capacity, which can be used as short circuit and overload protection for electric vehicle drive power system and other equipment.
注：本规范中所有试验数据均为直流环境下测得，测试条件和接线参考IEC 60269。
Note: All test data in this datasheet are measured in DC environment, test conditions and wiring refer to IEC 60269.

产品标识/ Product labeling

RS0ER-L2Q（200A-400A）





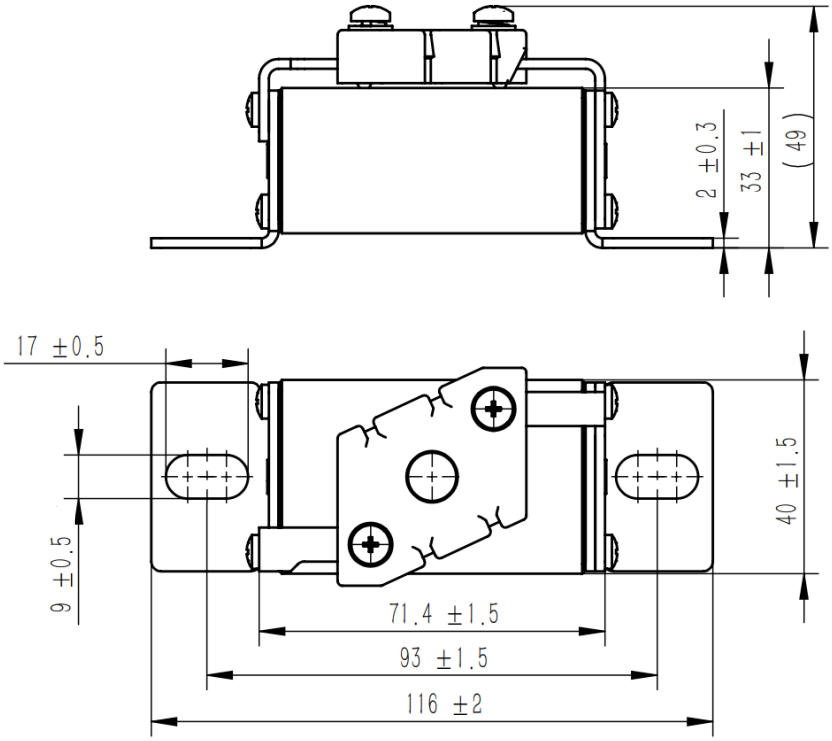
RS0ER-L2Q-N（200A-400A）





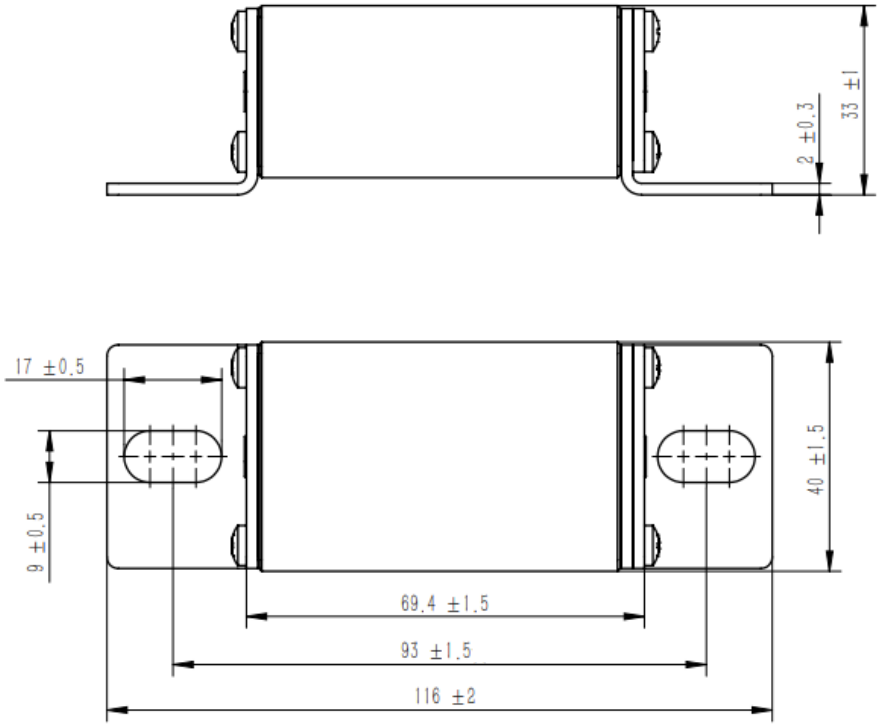
产品外形尺寸/Dimensions(mm)

RS0ER-L2Q（200A-400A）外形尺寸/Dimensions



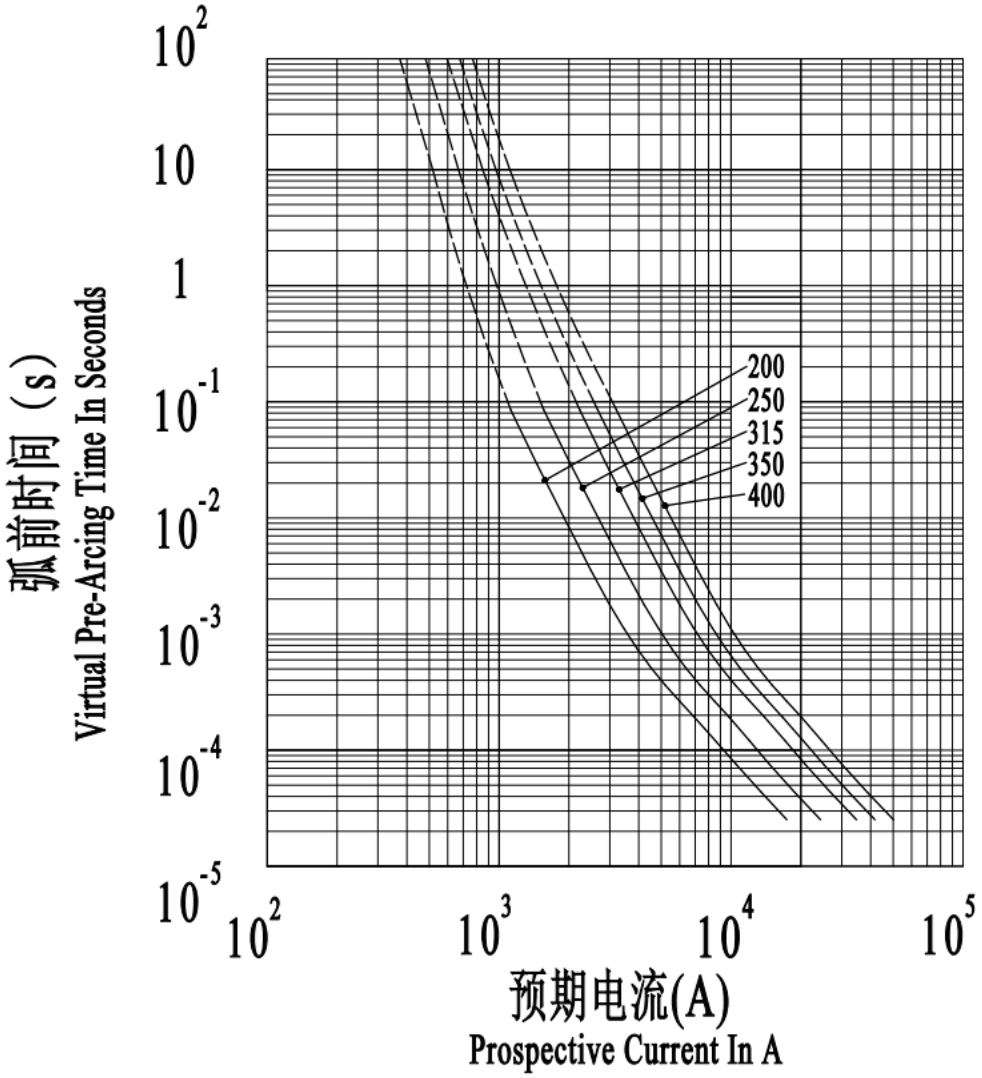
注：两触刀安装面共面度≤0.5。/Note: The coplanarity of the two contact knife mounting surfaces ≤ 0.5.

RS0ER-L2Q-N（200A-400A）外形尺寸/Dimensions



注：两触刀安装面共面度≤0.5。/Note: The coplanarity of the two contact knife mounting surfaces ≤ 0.5.

产品特性曲线/ Characteristic curve



注/Note：

1. 最小分断电流≥8In；/Minimum breaking current≥8In.
2. 特性曲线为电流方向误差±10%。/Characteristic curve for current direction error ±10%.

运输、存储/Transportation, Storage

运输/Transportation

避免过度挤压或冲击：在运输过程中，需要避免对熔断器施加过度挤压或直接冲击。这可能会损坏熔断器的外壳、接线端子或内部元件，从而影响其性能和安全性。运输过程中应避免雨雪侵袭和机械损伤。

Avoid excessive compression or shock: During transportation, it is necessary to avoid applying excessive compression or direct shock to the fuse. This may damage the fuse housing, terminals or internal components, thus affecting their performance and safety. Avoid rain, snow and mechanical damage during transportation.

防潮防湿：熔断器应远离潮湿和湿润的环境，以防止受潮和腐蚀。必要时，可以使用防潮包装材料或采取其他措施来保护熔断器免受潮湿影响。

Moisture and humidity protection: Fuses should be kept away from damp and wet environments to prevent moisture and corrosion. If necessary, moisture-proof packing materials may be used or other measures taken to protect the fuse from moisture.

存储/Storage

1.产品存储温度：-40℃～85℃；

Storage Temperature: -40℃～85℃.

2.包装存储温度：-40℃～70℃；

Package Temperature: -40℃～70℃.

3.相对湿度≤90%，无凝露。

Package Humidity: RH≤90%, no dewing.

使用条件/Use Conditions

大气条件/Atmosphere

空气是干净的，它的相对湿度在温度为40℃时不超过50%。

For clean air, maximum 50% RH at 40℃.

在较低温度下可以有较高的相对湿度，无明显凝露情况下，相对湿度可达95%。

Higher RH is allowed when temperature is lower, RH up to 95% without significant condensation.

周围空气温度/Ambient Air Temperature

标准典型测试周围空气温度：20℃；

Standard Testing Ambient Temperature: 20℃.

允许工作的周围空气温度：-40℃至125℃；

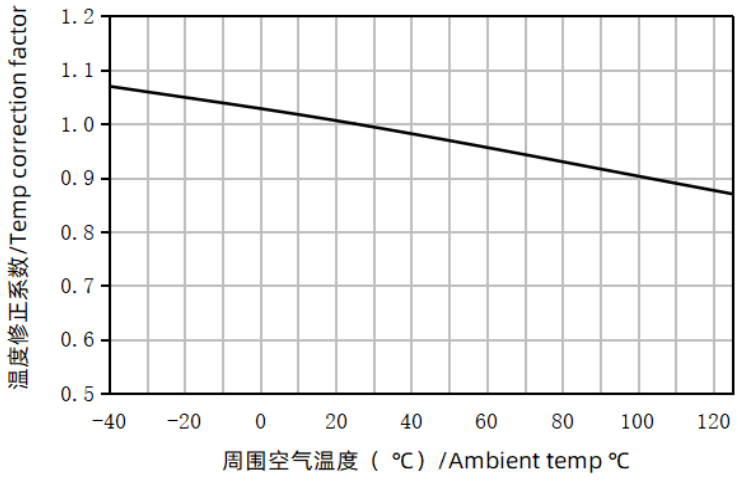
Tolerable Range: -40°～125℃.

当熔断器表面（壳体）温度工作至＞125℃时，经过评估或试验测试，双方达成共识后，熔断器也可被正常选用。

When the fuse surface (body) temperature >125℃, the fuse can also be selected normally after evaluation or pilot testing and mutual agreement.

熔断器在周围空气温度20℃以上工作，额定电流按照下面曲线进行修正。

When the fuse operates at an ambient air temperature above 20℃, the rated current shall be adjusted according to the following curve.



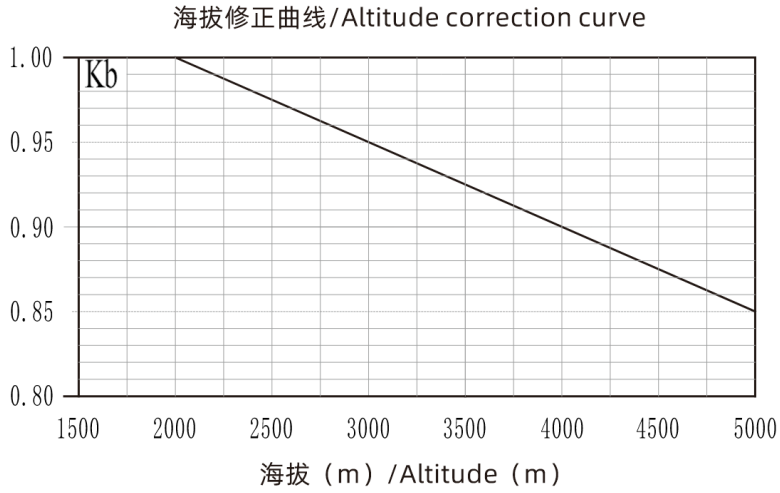
海拔/Altitude

标准典型测试条件：海拔≤2000m；

Standard Testing Conditions: Altitude ≤2000m.

允许使用条件：海拔≤5000m，不同海拔下熔断器额定电流按照下面曲线进行修正。

Tolerable Range: Altitude≤5000m；rated current should be corrected according to the following curve at different altitudes.



当海拔超过5000m，经过评估或试验测试，双方达成共识后，熔断器也可被正常选用。

When altitude above 5000m, fuses can also be selected normally after evaluation or pilot testing and mutual agreement.

安装条件/Mounting Condition

标准典型测试条件：无风，周围1m以内无发热器件；

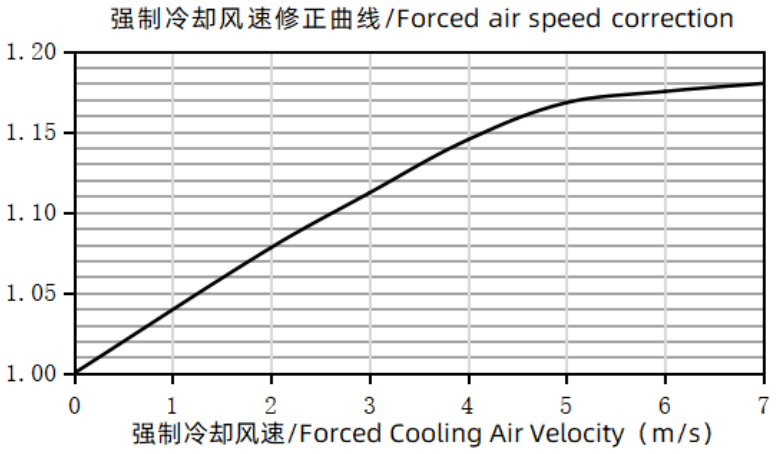
Standard Testing Condition: no ventilation, no heat source within 1m.

允许使用条件：允许有风及周围有发热器件，但需考虑其影响；

Tolerable Range: ventilation and heat source allowed, but their impacts should be considered.

强制风冷：采用强制风冷可以增加熔断器的散热，可以增加熔断器的额定电流；强制风冷条件下熔断器额定电流按照下面曲线进行修正。

Forced Air Cooling：The use of forced air cooling can enhance the heat dissipation of the fuse, thereby allowing an increase in its rated current. Rated current should be corrected according to the following curve.



铜导线/铜排尺寸/Copper wire/busbar size

标准典型测试条件：铜导线/铜排尺寸满足标准要求；

Standard Typical Testing Condition: The size of the copper wire/busbar complies with the standard requirements.

允许使用条件：推荐铜导体截面不小于标准截面的50%（标准截面以实际通流值为前提），其他材质导体需换算为铜导体的尺寸进行校核。

Tolerable Condition: It is recommended that the cross-sectional area of the copper conductor should not be less than 50% of the standard cross-sectional area (the standard cross-section is based on actual current-carrying capacity). For conductors made of other materials, their dimensions must be converted to equivalent copper conductor dimensions for verification.

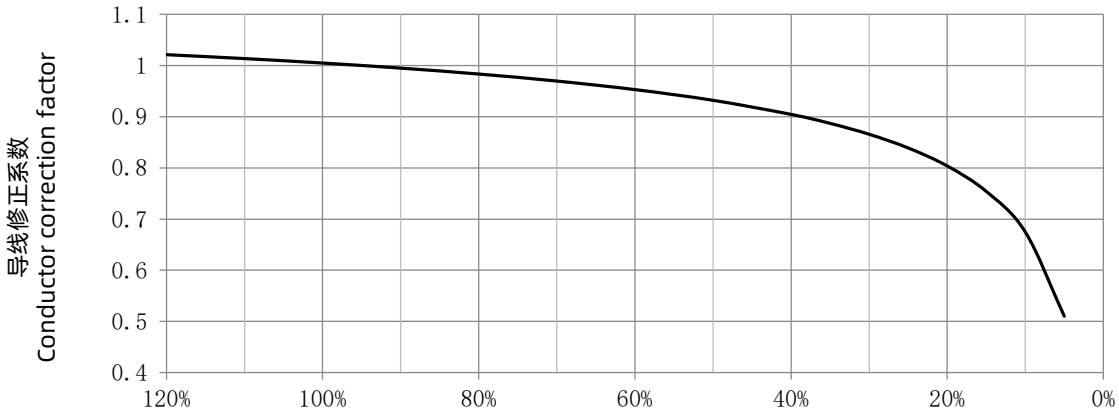
标准导线选用表（参考GB/T13539.1）/Standard Conductor Selection Chart（GB/T13539.1）

额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
2、4、6	1
8、10、12	1.5
16、20	2.5
25	4
32	6
40、50	10
63	16
80	25
100	35
125	50
160	70

额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
200、225	95
250、300	120
315、350	185
400、425	240
500	300
630	370
800	480
1000	600
1250	800
>1250	（1.0~1.6）A/mm²

不同铜导体尺寸下熔断器的额定电流按照下面曲线进行修正

Rated current should be corrected according to the following curve



振动环境及耐受地震等灾害能力/Vibration and Resistance to Disasters

振动满足GB 28046.3　4.1.2.4试验IV--乘用车弹性体(车身)要求;

Vibration meets GB 28046.3 4.1.2.4 Test IV.

机械冲击满足GB 28046.3　4.2.2;

Mechanical Shock meets GB 28046.3 4.2.2.

以上为基本条件，更为严酷的应用环境，请咨询我司。

For serious application condition, please consult Sinofuse.

污染等级/Pollution

圆管熔断器满足污染等级等级1（详见GB 14048.1 6.1.3.2）；
Cylindrical fuse meets pollution level Ⅰ (for details, see GB 14048.1 6.1.3.2).
方体熔断器满足污染等级等级2（详见GB 14048.1 6.1.3.2）。
Square body fuse meets pollution level Ⅱ (for details, see GB 14048.1 6.1.3.2).

电气间隙和爬电距离（以下简称绝缘尺寸）/Electrical Clearance and Creepage Distance (Hereafter Insulation Size)

绝缘尺寸受到多种因素的影响，包括海拔高度、环境温度、湿度、污染程度、材料组别、绝缘水平等。实际应用中绝缘尺寸的确定和校核参考GB/T 16935.1。

Insulation size is affected by many factors, including altitude, ambient temperature, humidity, pollution level, material group, insulation level and so on. In practical applications, the determination and verification of insulation dimensions should refer to GB/T 16935.1.

熔断器本体的绝缘尺寸一般远大于GB/T 16935.1标准要求的绝缘尺寸，除个别体积较小的熔断体外，无需校核熔断器本体绝缘尺寸。不同熔断器之间,以及熔断器对地的绝缘尺寸，需由用户进行校核。

The insulation size of the fuse is generally much larger than the insulation size required by GB/T 16935.1 standard, except for some smaller fuse links, there is no need to check the insulation size of the fuse link. The insulation size between different fuses and the fuse to ground should be checked by the user.

降低电压下的熔断I^2t值的计算/Calculation of Clearing I^2t Value at Reduced Voltage

使用下式可估算较低电压下的熔断I^2t值(参照GB/T13539.1):
The value of Clearing I^2t at lower voltages can be estimated using the following formula:

降低电压V_r，下的熔断 $I^2t=\{\frac{\text{试验电压}V_f\text{下的熔断}I^2t}{\text{弧前}I^2t}\}^{V_r/V_f}\times \text{弧前}I^2t$

Reduced Voltage V_r, Clearing $I^2t=\{\frac{\text{Clearing } I^2t \text{ at Test voltage } V_f}{Prearcing\ I^2t}\}^{V_r/V_f} x Prearcing\ I^2t$

注：上式仅用于单一测试参数下仅试验电压降低后熔断I^2t的估算；
Note: the above formula is only used for the estimation of clearing I^2t after only the test voltage is reduced for a single test parameter.

串联和并联使用说明/Series and Parallel Usage Instructions

熔断器并联时，施加的电压不得超过0.9Un（Un为熔断器额定电压，考虑到并联的各熔断器所承受的能量可能无法完全一致）；还应确保不超过各自在当前条件下允许的最大负载电流（考虑到并联不同熔断器电流分布可能不均以及熔断器间的热干涉）。

When fuses are connected in parallel, the applied voltage shall not exceed 0.9 Un (Un is the rated voltage of the fuses, taking into account that the energy absorbed by each parallel fuse may not be entirely consistent); it shall also be ensured that the maximum permissible load current of the respective fuses under the prevailing conditions is not exceeded (taking into account the possible uneven current distribution among parallel-connected fuses and the thermal interference between them).

熔断器串联时，建议串联的熔断器只在熔断时间小于10ms、且每个熔断器的恢复电压不超过0.9Un（Un为熔断器额定电压，考虑到串联的熔断器可能无法同时动作以均分施加的电压）。

When fuses are connected in series, it is recommended that the series-connected fuses are only connected when the clearing time is less than 10ms and the recovery voltage of each fuse does not exceed 0.9Un (Un is the rated voltage of the fuse, taking into account that the series-connected fuses may not be able to operate at the same time in order to equally share the applied voltage).

安装条件/Mounting Conditions

通用使用条件/General Usage Condition

1.参照产品规格书推荐扭矩进行安装，扭矩为动态扭矩，测试方法推荐拧紧法；

Install according to the recommended torque specified in the product specification. The torque is dynamic torque, and the tightening method is recommended for testing.

2.安装前，应检查熔断器的额定电压、额定电流、分断能力是否与要求的一致；

Before installation, check whether the rated voltage, rated current and breaking capacity of the fuse are the same as those required.

3.安装时，应保证熔断器端子及端子与对手件接触良好，建议对手件两端共面度≤熔断器要求的共面度且越小越好，安装面须可靠固定，避免安装面之间高度差造成产品触刀上下翘曲变形甚至机械损坏；

During installation, ensure good contact between the fuse terminals and their mating parts. It is recommended the coplanarity between the ends of the counterparts ≤ the coplanarity required by the fuse and the mounting surfaces should be reliably fixed to avoid warping and deformation of the product terminals up and down or mechanical damage.

4.安装时端子安装接触面无氧化、无杂质；

No oxidization and impurity on the contact surface of the terminal during installation.

5.安装过程中，禁止对熔断体本体敲击或撞击，防止对金属镀层的损坏；

During installation, it is forbidden to knock or impact on the fuse link to prevent damage to the metal plating.

6.安装时注意不使熔断器受到机械损伤；

Be careful not to subject the fuse to mechanical damage during installation.

7.熔断器更换必须在不带电的情况下进行，禁止带电插拔/拆装；

Fuse replacement must be carried out in the case of non-electricity.

8.熔断器位于封闭环境时，应增加额外的辅助散热措施或适当增加额定电流；

When the fuse is located in a closed environment, there should be additional auxiliary heat dissipation measures or the appropriate increase in the rated current.

9.熔断器的连接线的材料和截面积应符合选型约定，不得随意更改，以免发生误动作或其他异常；

The materials and cross-sectional area of the connecting wires of the fuse should be in accordance with the selection agreement, and shall not be changed arbitrarily to avoid false operation or other abnormalities.

10、熔断器应与其他发热器件或热敏感器件保持足够的间距；

The fuse should be with other heat devices or heat-sensitive devices to maintain adequate spacing.

11.对于内螺纹连接式熔断体安装要求及扭矩参考规格书推荐，若使用时需更改推荐参数，需双方协商约定；

For internally threaded connection-type fuse links, the installation requirements and torque values should follow the recommendations provided in the Specification. If any changes to the recommended parameters are required during use, they must be agreed upon by both parties through consultation.

12.螺栓安装式熔断器，拆装次数≤5次。

For bolted fuses, the number of disassembly cycles should not exceed 5.

安装条件/Mounting Conditions

PCB焊接型结构/PCB Welded Structure

1.熔断器应与其他发热器件或热敏感器件保持足够的间距，若其1m范围内有其他发热部件或散热部件，需要考虑对熔断器的热效应并进行相应降额；

Fuses should be sufficiently spaced from other heat-generating or heat-sensitive devices, and if there are other heat-generating or heat-sinking parts within 1m, the thermal effect on the fuse needs to be considered and derated accordingly.

2.若熔断器位于封闭环境时，应增加额外的辅助散热措施或适当增加额定电流；

Additional auxiliary heat dissipation measures should be added or the rated current should be increased appropriately if the fuse is located in an enclosed environment.

3.需保证安装后的产品可靠固定，焊料需完全包裹引脚片，确保可靠连接，不允许虚焊；

After installation, the product should be fixed reliably, and the solder should be completely wrapped around the pin pieces to ensure a reliable connection without false soldering.

4.产品直接焊接在PCB板上，避免产品大幅度倾斜导致焊接不良；

The product is soldered directly to the PCB board to avoid the product tilted significantly resulting in poor soldering.

5.应保证引脚与PCB板子良好电接触，避免镀层氧化或引脚与PCB板间有其他杂质，直接接触面积≥引脚横截面积；

It should be ensured that the pins have good electrical contact with the PCB, avoiding oxidation of the plating or the presence of any impurities between the pins and the PCB. The direct contact area should be no less than the cross-sectional area of the pins.

6.PCB板安装孔间距和熔断器安装尺寸一致，且需要同轴，避免安装、使用过程中不适配强行装入，导致产品引脚损坏；

The spacing of the mounting holes on the PCB should match the installation dimensions of the fuse and be coaxial to avoid forced fitting during installation or use, which could lead to damage of the product pins.

7.产品直接焊接在PCB上，可垂直、水平或倾斜安装；

The product is welded directly to the PCB and can be installed vertically, horizontally or at an angle.

8.波峰焊接温度建议在260℃以下，焊接时间10秒以下，建议提前预热后再进行焊接，焊接时避免强烈震动和显著冲击详细参见规格书要求；

Wave soldering temperature is recommended below 260 ° C, welding time of 10 seconds or less; it is recommended to preheat before soldering, and to avoid strong vibrations and significant impacts during the soldering process. For detailed requirements, please refer to the Specification.

9.产品须可靠焊接，避免虚焊影响产品性能，引脚处应呈现至少95%区域的连续均匀的焊料涂层；

The product must be reliably soldered to avoid cold joints that could impair its performance. A continuous and uniform solder coating should cover at least 95% of the pin area.

10.保持熔断器周边与PCB板子内元器件有一定的间距，具体距离要求满足电气间隙爬电距离的要求；

Maintain a certain spacing between the fuse and the components on the PCB, and ensure that the specific distance meets the requirements for electrical clearance and creepage distance.

11.熔断器和PCB板子的连接须考虑锡须的影响，避免造成产品的短路失效。

The connection between the fuse and the PCB must take into account the impact of tin whiskers to avoid causing short-circuit failures in the product.

圆帽型结构/Cylinder Cap Structure

对于圆筒帽型产品需采用夹子或底座可靠固定安装后使用，安装条件需关注振动、冲击工况要求；

For cylindrical cap-type products, they must be securely installed using clips or a base before use, and the installation conditions should take into account the requirements for vibration and shock conditions.